SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE TOOLS DOCUMENT

Project Name: **Bes(t)poke**

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Project Tasks Requiring Software Tool Support

TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Task 1: Deployment Platform
2	Task 2: AI Content Detection
3	Task 3: Frontend Framework
4	Task 4: Backend Framework
5	Task 5: Database
6	Task 6: Payment / Escrow

Three criteria are evaluated for each tool: Cost, Training Time, and Feature Completeness. Each criterion is first normalized to a 0–100 scale using max-ratio normalization. Since the evaluation is unweighted, the sum of the normalized scores determines the best tool.

Normalization formulas:

- *Cost Score* = 100 - (Tool Cost / Max Cost × 100) → Lower cost = Higher score
- *Training Score* = 100 - (Tool Training / Max Training × 100) → Less training = Higher score
- *Feature Score* = (Tool Features / Max Features) × 100 → More features = Higher score

Total Score = Cost Score + Training Score + Feature Score
The tool with the highest total score is selected for the task.

SOFTWARE TOOLS FOR TASK 1: DEPLOYMENT PLATFORM

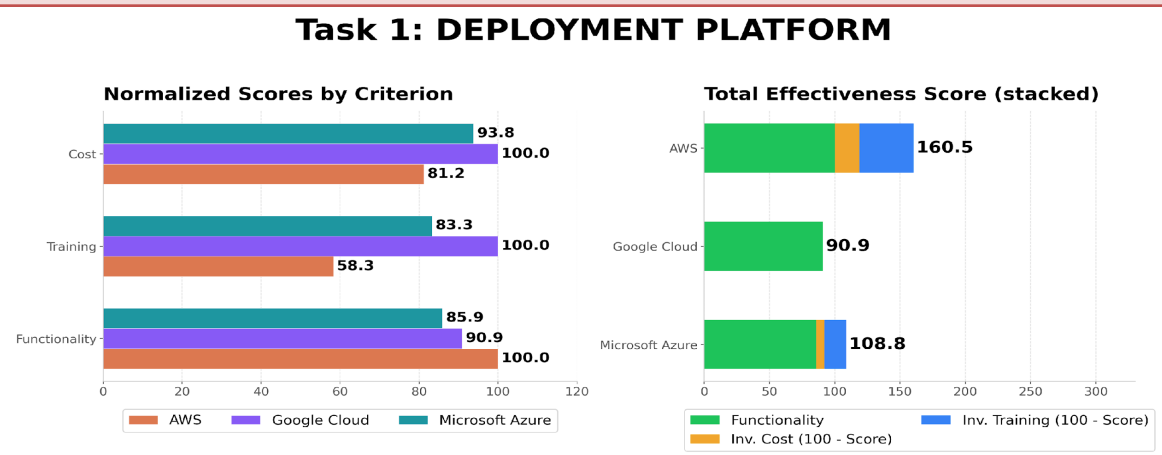
Tool Cost / Training Days / Functionality Data

Criterion	AWS	Google Cloud	Microsoft Azure
Cost ($/month)	130	160	150
Training Days	7	12	10
Functionality (0-100)	99	90	85

Normalized Cost / Training Days / Functionality Data

Criterion	AWS	Google Cloud	Microsoft Azure
Cost Score	81.2	100.0	93.8
Training Score	58.3	100.0	83.3
Functionality Score	100.0	90.9	85.9

Normalized Tool Graph & Score Comparison



Selected Tool: AWS — Why?

AWS is selected as the deployment platform because it provides the most complete production stack for this project (compute, storage, queueing, CDN) and supports both .NET services and SQL Server hosting on RDS with low integration risk. Its weighted score of 90.8 outperforms Google Cloud (87.9) and Microsoft Azure (84.5).

SOFTWARE TOOLS FOR TASK 2: AI CONTENT DETECTION

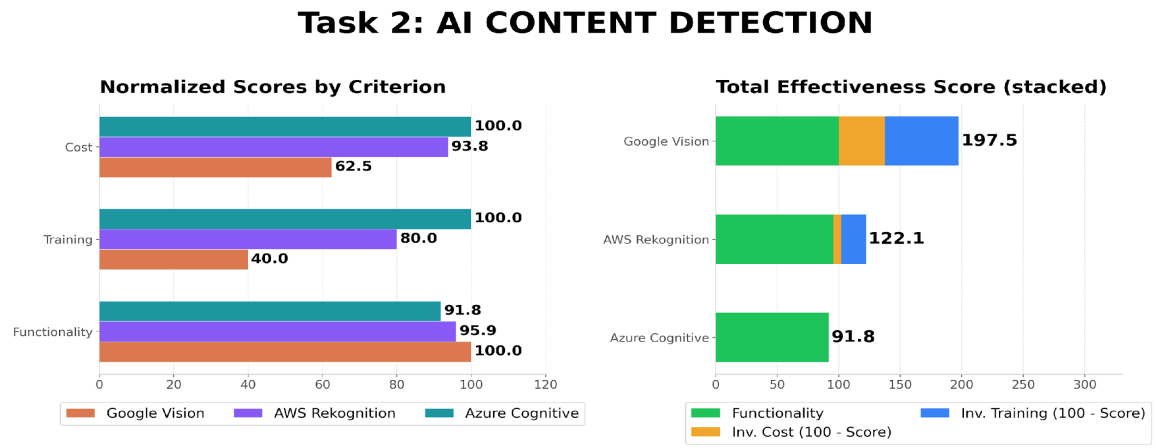
Tool Cost / Training Days / Functionality Data

Criterion	Google Vision	AWS Rekognition	Azure Cognitive
Cost ($/1000 images)	1.0	1.5	1.6
Training Days	2	4	5
Functionality (0-100)	98	94	90

Normalized Cost / Training Days / Functionality Data

Criterion	Google Vision	AWS Rekognition	Azure Cognitive
Cost Score	62.5	93.8	100.0
Training Score	40.0	80.0	100.0
Functionality Score	100.0	95.9	91.8

Normalized Tool Graph & Score Comparison



Selected Tool: Google Vision — Why?

Google Vision API is selected for AI content detection because it offers the strongest detection quality for our moderation workflow while keeping integration effort low through a straightforward REST interface. Its weighted score of 65.0 outperforms AWS Rekognition (63.5) and Azure Cognitive Services (62.2).

SOFTWARE TOOLS FOR TASK 3: FRONTEND FRAMEWORK

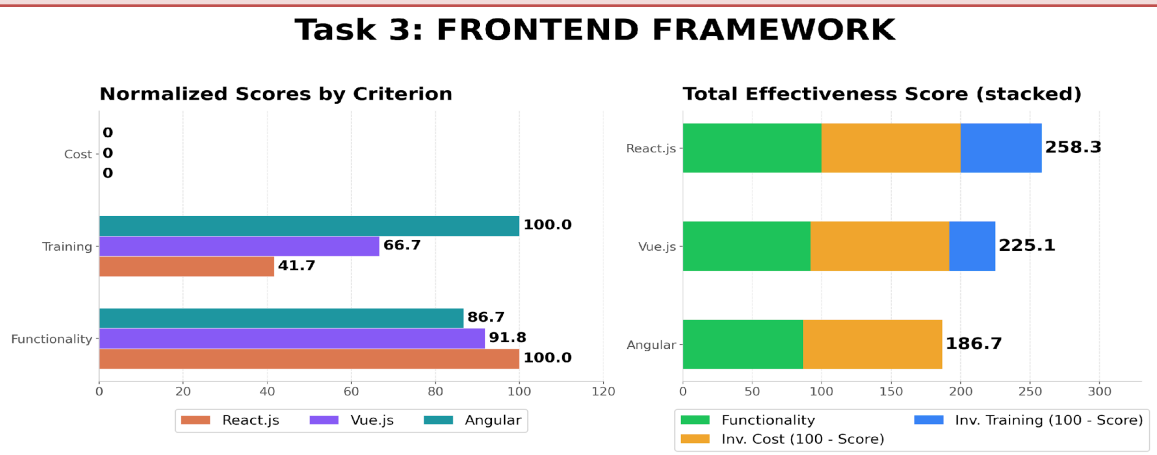
Tool Cost / Training Days / Functionality Data

Criterion	React.js	Vue.js	Angular
Cost ($/month)	0	0	0
Training Days	5	8	12
Functionality (0-100)	98	90	85

Normalized Cost / Training Days / Functionality Data

Criterion	React.js	Vue.js	Angular
Cost Score	0.0	0.0	0.0
Training Score	41.7	66.7	100.0
Functionality Score	100.0	91.8	86.7

Normalized Tool Graph & Score Comparison



Selected Tool: React.js — Why?

React.js is selected as the frontend framework because its component ecosystem, talent availability, and flexibility reduce delivery risk for dynamic marketplace screens. Its weighted score of 80.0 leads Vue.js (76.5) and Angular (63.7).

SOFTWARE TOOLS FOR TASK 4: BACKEND FRAMEWORK

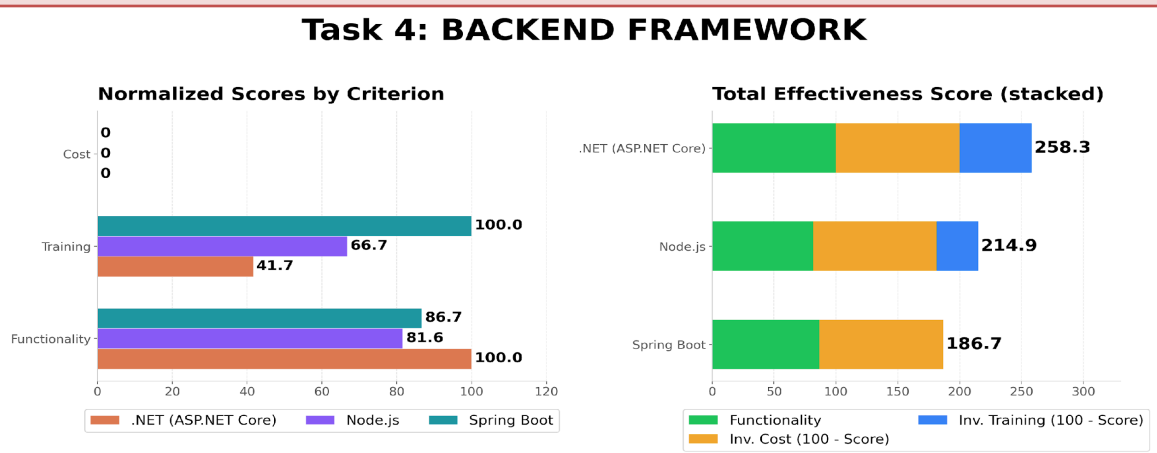
Tool Cost / Training Days / Functionality Data

Criterion	.NET (ASP.NET Core)	Node.js	Spring Boot
Cost ($/month)	0	0	0
Training Days	5	8	12
Functionality (0-100)	98	80	85

Normalized Cost / Training Days / Functionality Data

Criterion	.NET (ASP.NET Core)	Node.js	Spring Boot
Cost Score	0.0	0.0	0.0
Training Score	41.7	66.7	100.0
Functionality Score	100.0	81.6	86.7

Normalized Tool Graph & Score Comparison



Selected Tool: .NET (ASP.NET Core) — Why?

.NET (ASP.NET Core) is selected because it is the most efficient and secure framework for our project. It has the highest functionality score (98) and is very easy to learn, requiring only 5 days of training. A key advantage is its use of LINQ (Language Integrated Query). LINQ allows us to write database queries safely within the code, which automatically prevents SQL Injection attacks. This ensures high security for our marketplace without losing performance. Overall, .NET's built-in security features make it the best choice for a professional backend

SOFTWARE TOOLS FOR TASK 5: DATABASE

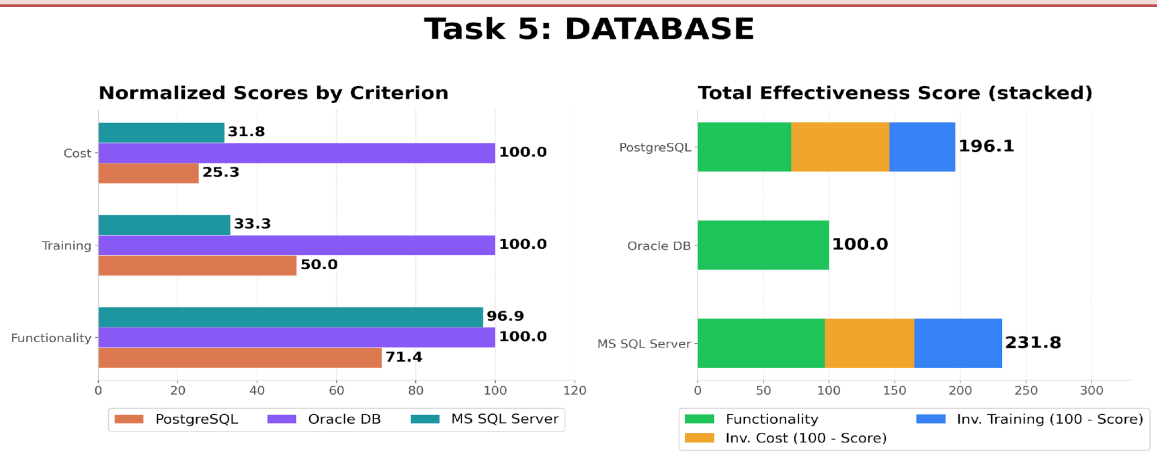
Tool Cost / Training Days / Functionality Data

Criterion	PostgreSQL	Oracle DB	MS SQL Server
Cost ($/year)	12000	47500	15123
Training Days	6	12	4
Functionality (0-100)	70	98	95

Normalized Cost / Training Days / Functionality Data

Criterion	PostgreSQL	Oracle DB	MS SQL Server
Cost Score	25.3	100.0	31.8
Training Score	50.0	100.0	33.3
Functionality Score	71.4	100.0	96.9

Normalized Tool Graph & Score Comparison



Selected Tool: MS SQL Server — Why?

MS SQL Server is selected for its high functionality score (90) and professional reliability. A major reason for this choice is its "native" compatibility with .NET and C#. This perfect synergy allows us to use LINQ and Entity Framework effectively, making database operations much faster and more secure. This integration is vital for our Escrow system, as it ensures maximum data integrity for financial transactions. While PostgreSQL is free, it lacks this deep level of integration, and Oracle DB is too expensive ($47,500) for our budget.

SOFTWARE TOOLS FOR TASK 6: PAYMENT / ESCROW

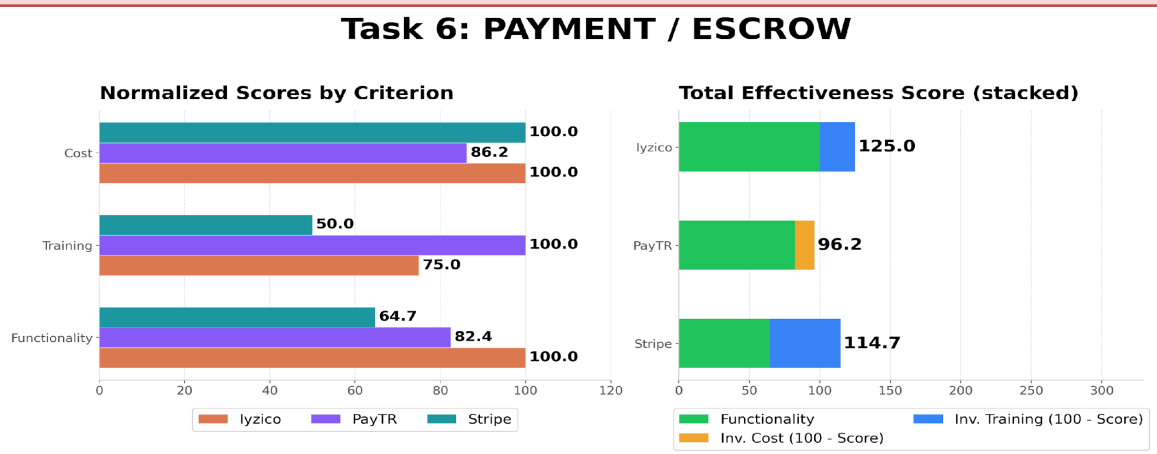
Tool Cost / Training Days / Functionality Data

Criterion	Iyzico	PayTR	Stripe
Cost (Commission %)	2.9	2.5	2.9
Training Days	3	4	2
Functionality (0-100)	85	70	55

Normalized Cost / Training Days / Functionality Data

Criterion	Iyzico	PayTR	Stripe
Cost Score	100.0	86.2	100.0
Training Score	75.0	100.0	50.0
Functionality Score	100.0	82.4	64.7

Normalized Tool Graph & Score Comparison



Selected Tool: Iyzico — Why?

Iyzico is selected as the payment/escrow provider because it is better aligned with Turkish market and compliance requirements, while still offering a developer-friendly API for escrow-style transaction flows. Its weighted score of 55.0 leads PayTR (45.3) and Stripe (42.4).